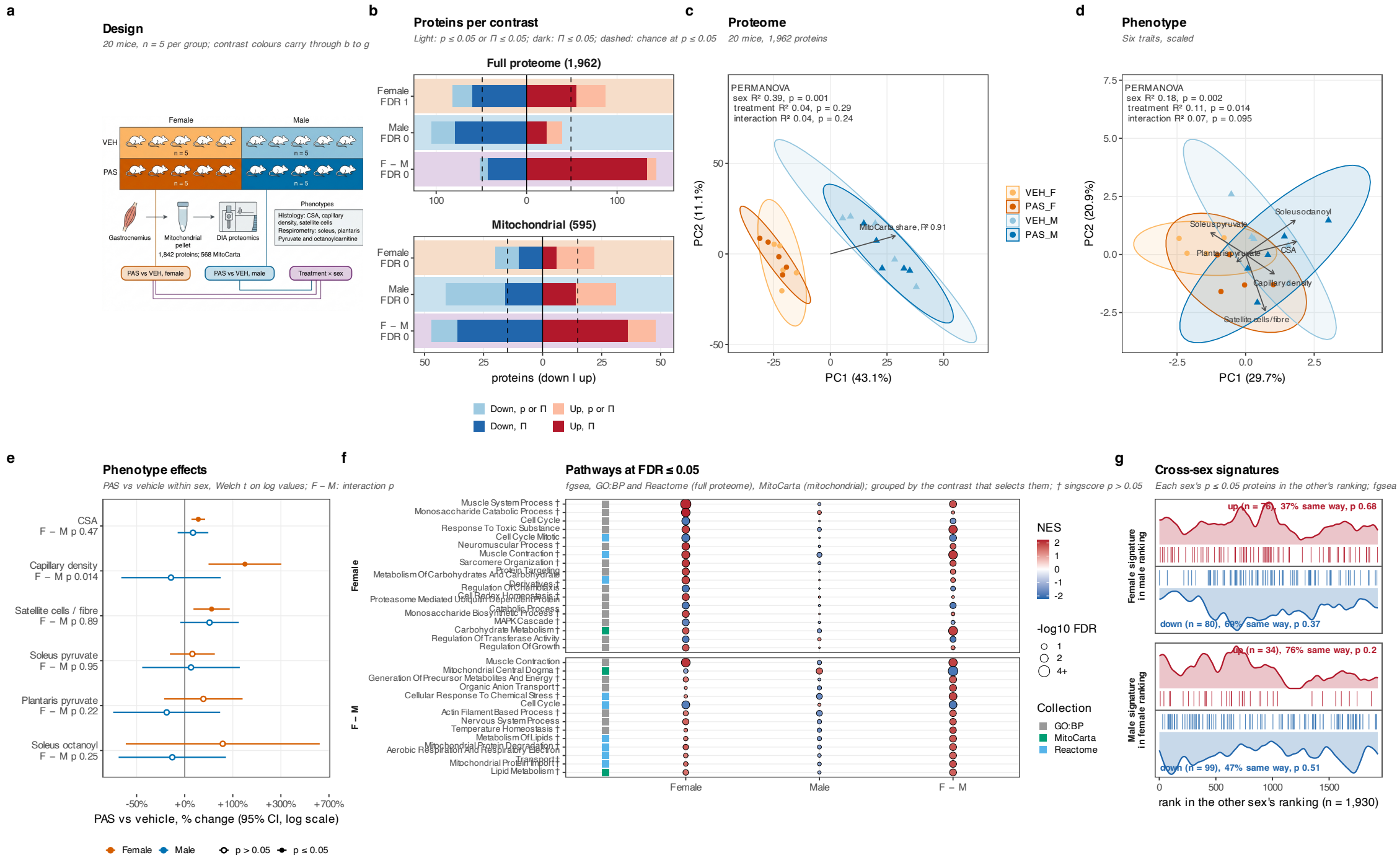

a
Design
20 mice, n = 5 per group; contrast colours carry through b to g
Female
Male
VEH
PAS
n = 5
Gastrocnemius
Mitochondrial pellet
DIA proteomics
1,842 proteins; 568 MitoCarta
Phenotypes
Histology: CSA, capillary density, satellite cells
Respirometry: soleus, plantaris
Pyruvate and octanoylcarnitine
PAS vs VEH, female
PAS vs VEH, male
Treatment × sex
b
Proteins per contrast
Light: p ≤ 0.05 or Π ≤ 0.05; dark: Π ≤ 0.05; dashed: chance at p ≤ 0.05
Full proteome (1,962)
Female FDR 1
Male FDR 0
F − M FDR 0
Mitochondrial (595)
proteins (down | up)
Down, p or Π
Up, p or Π
Down, Π
Up, Π
c
Proteome
20 mice, 1,962 proteins
PERMANOVA
sex R² 0.39, p = 0.001
treatment R² 0.04, p = 0.29
interaction R² 0.04, p = 0.24
MitoCarta share, R² 0.91
PC1 (43.1%)
PC2 (11.1%)
VEH_F
PAS_F
VEH_M
PAS_M
d
Phenotype
Six traits, scaled
PERMANOVA
sex R² 0.18, p = 0.002
treatment R² 0.11, p = 0.014
interaction R² 0.07, p = 0.095
PC1 (29.7%)
PC2 (20.9%)
Soleus octanoyl
Soleus pyruvate
Plantaris pyruvate
CSA
Capillary density
Satellite cells / fibre
e
Phenotype effects
PAS vs vehicle within sex, Welch t on log values; F − M: interaction p
CSA
F − M p 0.47
Capillary density
F − M p 0.014
Satellite cells / fibre
F − M p 0.89
Soleus pyruvate
F − M p 0.95
Plantaris pyruvate
F − M p 0.22
Soleus octanoyl
F − M p 0.25
-50%
+0%
+100%
+300%
+700%
PAS vs vehicle, % change (95% CI, log scale)
Female
Male
p > 0.05
p ≤ 0.05
f
Pathways at FDR ≤ 0.05
fgsea, GO:BP and Reactome (full proteome), MitoCarta (mitochondrial); grouped by the contrast that selects them; † singscore p > 0.05
Female
F − M
Muscle System Process †
Monosaccharide Catabolic Process †
Cell Cycle
Response To Toxic Substance
Cell Cycle Mitotic
Neuromuscular Process †
Muscle Contraction †
Sarcomere Organization †
Protein Targeting
Metabolism Of Carbohydrates And Carbohydrate Derivatives †
Regulation Of Chemotaxis
Cell Redox Homeostasis †
Proteasome Mediated Ubiquitin Dependent Protein Catabolic Process
Monosaccharide Biosynthetic Process †
MAPK Cascade †
Carbohydrate Metabolism †
Regulation Of Transferase Activity
Regulation Of Growth
Muscle Contraction
Mitochondrial Central Dogma †
Generation Of Precursor Metabolites And Energy †
Organic Anion Transport †
Cellular Response To Chemical Stress †
Cell Cycle
Actin Filament Based Process †
Nervous System Process
Temperature Homeostasis †
Metabolism Of Lipids †
Mitochondrial Protein Degradation †
Aerobic Respiration And Respiratory Electron Transport †
Mitochondrial Protein Import †
Lipid Metabolism †
Female
Male
F − M
NES
-log10 FDR
Collection
GO:BP
MitoCarta
Reactome
g
Cross-sex signatures
Each sex's p ≤ 0.05 proteins in the other's ranking; fgsea
Female signature in male ranking
up (n = 76), 37% same way, p 0.68
down (n = 80), 69% same way, p 0.37
Male signature in female ranking
up (n = 34), 76% same way, p 0.2
down (n = 99), 47% same way, p 0.51
rank in the other sex's ranking (n = 1,930)